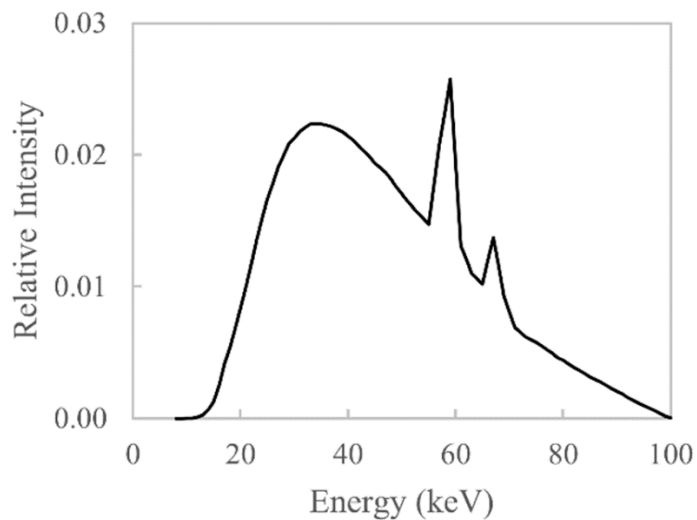


X-Ray, CT, and PET/SPECT

Q1: X-Ray (5 Points)

Below is a typical X-ray spectrum produced by an X-ray tube. The spectrum shows the intensity of X-rays (y-axis) as a function of their energy (x-axis).

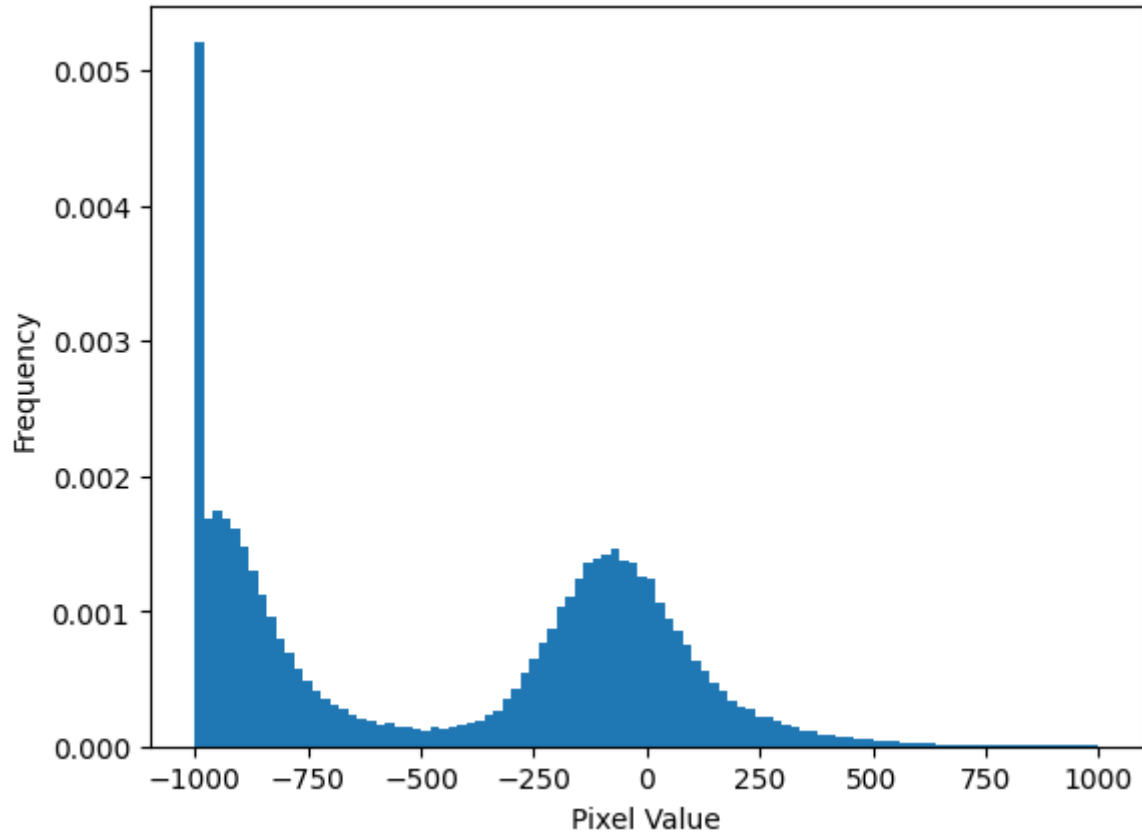
- Identify and label the regions corresponding to bremsstrahlung radiation (1 Point)
- Identify and label the peaks corresponding to characteristic radiation (1 Point)



- Provide a brief explanation of how each type of radiation is produced and why they appear differently in the spectrum (3 Points):

Q2: Windowing (5 Points)

Below is a histogram showing the distribution of intensity values (in Hounsfield Units) from a CT image of a patient's chest. The x-axis represents the Hounsfield Units (HUs), and the y-axis represents the frequency of each HU value.



- Define a suitable window to best visualize the soft tissue (values around zero). Specify the minimum and maximum Hounsfield Units (HUs) for this window (2 Points)

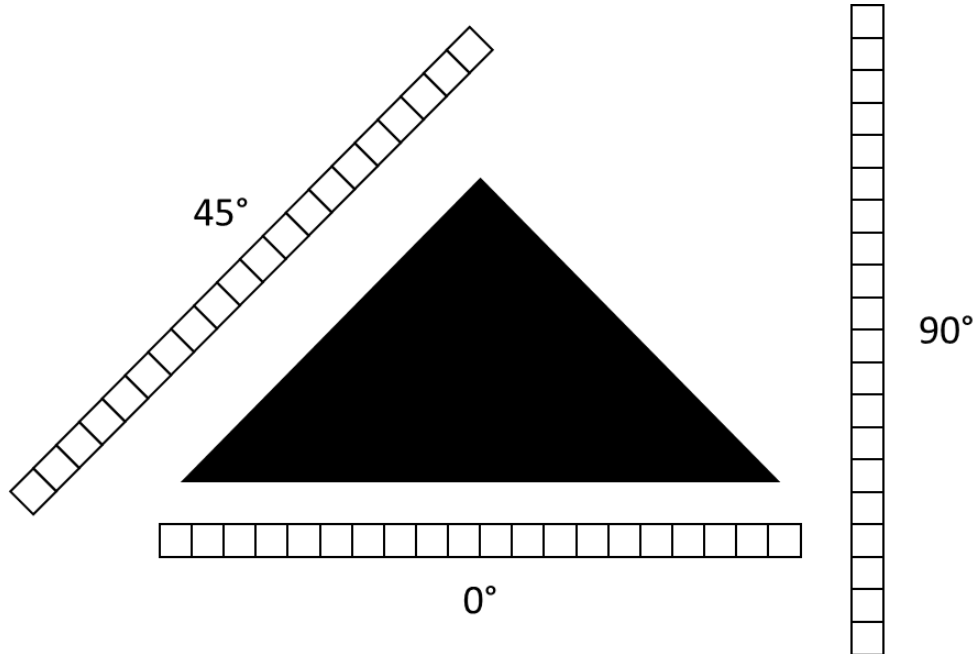
minimum: _____ maximum: _____

- Explain your reasoning for selecting this window. Discuss how the chosen window settings will help in enhancing the visualization of soft tissue and differentiate it from other structures (such as the lung) (3 Points):

Q3: CT (5 Points)

Below is the depiction of a triangle that will be scanned using a virtual CT scanner. The task is to sketch the projection of this triangle at three different angles (0° , 45° , and 90°).

- Sketch the projection of the triangle at the following angles: 0° , 45° , and 90° . For each angle, qualitatively shade the corresponding projection below (2 Points)



- What do you call a matrix that consists of these projections as columns? (1 Point):
- Describe how the triangle could be reconstructed from these projections (2 Points):

Q4: SPECT vs PET (3 Points)

What is the main difference between SPECT and PET?